Junjie (Takku) Li

@ alvinbluy@gmail.com | Portfolio | LinkedIn | GitHub | Seattle (Open to Relocate)

Education

Northeastern University Seattle, WA | Sep 2023 - Dec 2025
Master of Science in Computer Science (GPA: 3.9/4.0)
Teaching Assistant: Object Oriented Design (Spring 2024)

Nankai University Tianjin, China
B.S. in Mathematics

Experience

Berkshire Hathaway HomeServices: Intern Seattle, WA | Jun 2024 - Aug 2024
Machine Learning Engineer
- Built data pipelines using **Airflow** to automate **ETL** processes, reducing data processing time by 17%.
- Customized and fine-tuned **deep learning models** (N-Beats series) for housing price and inventory predictions, improving model accuracy by 30%. Applied **Agile** principles to iteratively refine data acquisition processes and update models, resulting in a 25% reduction in development cycle time.

Temple University: PhD Research Assistant Philadelphia, PA | Sep 2021 - May 2023
Performed text analysis on streaming comments using NLP techniques.
- Preprocessed live comment data with **SQL** for extraction, followed by further analysis using **Pandas** and **NumPy**. Extracted features from images and text with **ResNet-50** and pre-trained **BERT** embeddings.
- Improved data processing scalability and efficiency by 5x using **Dask** and **NVIDIA RAPIDS** suite.

Projects

Search Engine for Emojis | *Link* LLM-RAG Python/Go Web
"Emoji Search", a RAG-based AI search engine finding emojis based on web services.
- Adopted an event-driven architecture (**EDA**) for smoother search experience with four main components.
- Streamed user search queries through **Kafka** and integrated **Spark** to process large volumes of streaming data, improving real-time query handling by 20% and reducing batch processing time by 35% for analytics.
- Optimized microservice discovery and balanced queries, resulting in 30% fewer service failures and 15% faster response times at 5000 QPS. Containerized it in **Docker** and deployed it on **AWS EKS**.

AsyncTask Scheduler | Collaborated with *Hui Zhou* | *Link* Distributed Go Framework
"Async4Ai", a light-weighted framework asynchronously processing interdependent tasks.
- Designed a distributed task-scheduling system that decouples producers from consumers, highlighting complete task management via web module with **RESTful API** for interaction between agents and database.
- Crafted loosely coupled table normalization & sharding logic in **MySQL** to improve database performance.
- Performed stress testing and improved performance from 500 QPS to 2000 QPS by designing an efficient caching mechanism with **Redis**, using distributed locks and reconfiguring the database connection pool.

Anonymous Academic Career Hub | *Link* Full-stack JavaScript Web
"Business Job Market Rumors", a discussion board for academic jobs in business schools.
- Utilized **Express/Node** + **React** in a client-server model, integrating **Tailwind CSS**.
- Implemented role-based access control, context-based authentication using **JWT**, and automated content moderation using **Flask** to ensure a secure and respectful discussion environment.
- Deployed on **Azure App Service** via **GitHub Actions** with **MongoDB** hosted on **Azure Cosmos DB**.

TECHNICAL SKILLS

Languages & Databases: Python(5yr), Java(2yr), Golang(2yr), C/C++(1yr), JavaScript, SQL; MongoDB, Redis
Frameworks: CUDA, Spark, Dask, PyTorch, TensorFlow, Langchain, Streamlit, Express, Node, Angular, FastAPI
Tools: Dataiku, NeMo, Flask, Gradle, JUnit, Postman, GitHub Actions(CI/CD), Docker, Spark, Kubernetes
Miscellaneous: Linux, AWS(*Certified Solutions Architect Associate*), Azure (*Certified AI Engineer Associate*)